

## Tarea 2.3

$$1) \lim_{x \rightarrow 2} f(x) = 4$$

$$\lim_{x \rightarrow 2} g(x) = -2$$

$$\lim_{x \rightarrow 2} h(x) = 0$$

$$a) \lim_{x \rightarrow 2} [f(x) + 5g(x)] = \lim_{x \rightarrow 2} f(x) + 5 \lim_{x \rightarrow 2} g(x) = 4 + 5(-2) = -6$$

$$b) \lim_{x \rightarrow 2} [g(x)]^3 = (-2)^3 = -8$$

$$c) \lim_{x \rightarrow 2} \sqrt{f(x)} = \sqrt{4} = 2$$

$$d) \lim_{x \rightarrow 2} \frac{3f(x)}{g(x)} = \frac{3 \lim_{x \rightarrow 2} f(x)}{\lim_{x \rightarrow 2} g(x)} = \frac{3(4)}{-2} = \frac{12}{-2} = -6$$

$$e) \lim_{x \rightarrow 2} \frac{g(x)}{h(x)} = \frac{\lim_{x \rightarrow 2} g(x)}{\lim_{x \rightarrow 2} h(x)} = \frac{-2}{0} = \text{No existe}$$

$$f) \lim_{x \rightarrow 2} \frac{g(x)h(x)}{f(x)} = \frac{-2(0)}{4} = \frac{0}{4} = 0$$

$$7) \lim_{x \rightarrow 8} (1 + \sqrt[3]{x})(2 - 6x^2 + x^3) = (1 + \sqrt[3]{8})(2 - 6(8)^2 + 8^3) = 3(130) = 390$$

$$9) \lim_{x \rightarrow 2} \sqrt{\frac{2x^2 + 1}{3x - 2}} = \sqrt{\frac{2(2)^2 + 1}{3(2) - 2}} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

$$17) \lim_{h \rightarrow 0} \frac{(-5+h)^2 - 25}{h} = \frac{(-5+0)^2 - 25}{0} = \frac{0}{0} \text{ Forma indeterminada}$$

$$\lim_{h \rightarrow 0} \frac{h^2 - 10h + 25 - 25}{h} = \frac{h(h-10)}{h} = 0 - 10 = -10$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{9+h} - 3}{h} = \frac{\sqrt{9+0} - 3}{0} = \frac{0}{0} \text{ forma indeterminada}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{9+h} - 3}{h} \cdot \frac{\sqrt{9+h} + 3}{\sqrt{9+h} + 3} = \frac{9+h-9}{h(\sqrt{9+h}+3)} = \frac{h}{h(\sqrt{9+h}+3)} = \frac{1}{\sqrt{9+h}+3}$$

$$\lim_{x \rightarrow 0} \frac{1}{\sqrt{9+h}+3} = \frac{1}{\sqrt{9+0}+3} = \frac{1}{6}$$

$$29) \lim_{t \rightarrow 0} \left( \frac{1}{t\sqrt{7+t}} - \frac{1}{t} \right) = \frac{1}{0\sqrt{7+0}} - \frac{1}{0} = \frac{0}{0}$$

$$\lim_{t \rightarrow 0} \left( \frac{1}{t\sqrt{7+t}} - \frac{1}{t} \right) = \frac{1-\sqrt{7+t}}{t\sqrt{7+t}} \cdot \frac{1+\sqrt{7+t}}{1+\sqrt{7+t}} = \frac{1-7-t}{t\sqrt{7+t}(1+\sqrt{7+t})} = \frac{-6-t}{t\sqrt{7+t}(1+\sqrt{7+t})} = \frac{-7}{\sqrt{7+7}(1+\sqrt{7+7})} = \frac{-7}{\sqrt{14}(1+\sqrt{14})} = -\frac{1}{2}$$

$$37) \text{ Si } 4x-9 \leq f(x) \leq x^2-4x+7 \text{ para } x \geq 0, \text{ encuentre } \lim_{x \rightarrow 4} f(x)$$

$$\lim_{x \rightarrow 4} 4x-9 \leq \lim_{x \rightarrow 4} f(x) \leq \lim_{x \rightarrow 4} x^2-4x+7$$

$$7 \leq \lim_{x \rightarrow 4} f(x) \leq 7$$

$$\lim_{x \rightarrow 4} f(x) = 7$$

$$39) \text{ Demuestra que } \lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} = 0$$

$$x^4 \cdot -1 \leq x^4 \cos \frac{2}{x} \leq x^4 \cdot 1$$

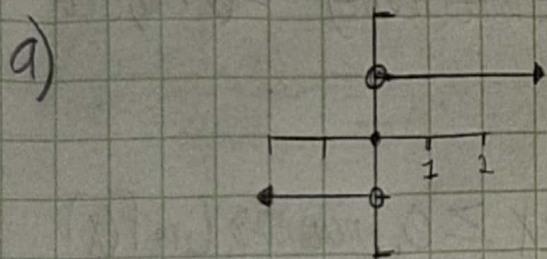
$$\lim_{x \rightarrow 0} -1 = \lim_{x \rightarrow 0} x^4 \cdot 1$$

$$0 = 0$$

$$43) \lim_{x \rightarrow 0.5} \frac{2x - 1}{12x^3 - x^2} = \frac{2(0.5) - 1}{12(0.5)^3 - 0.5^2} = \frac{0}{0}$$

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0.5} \frac{2}{-(6x^2 + 2x)} = \frac{2}{-7} = -\frac{2}{7}$$

$$47) \operatorname{sgn} x = \begin{cases} -1 & \text{si } x < 0 \\ 0 & \text{si } x = 0 \\ 1 & \text{si } x > 0 \end{cases}$$



$$b) (i) \lim_{x \rightarrow 0^+} \operatorname{sgn} x = 1$$

$$(ii) \lim_{x \rightarrow 0^-} \operatorname{sgn} x = -1$$

$$(iii) \lim_{x \rightarrow 0} \operatorname{sgn} x = \text{No existe}$$

$$(iv) \lim_{x \rightarrow 0} |\operatorname{sgn} x| = 1$$

$$51) \text{ Sea } B(t) = \begin{cases} 4 - \frac{1}{2}t & \text{si } t < 2 \\ \sqrt{t} + c & \text{si } t \geq 2 \end{cases}$$

Determinar el valor  $c$  tal que  $\lim_{t \rightarrow 2} B(t)$  exista

$$\lim_{t \rightarrow 2} 4 - \frac{1}{2}t$$

$$\lim_{t \rightarrow 2} \frac{8 - t}{2}$$

$$\lim_{t \rightarrow 2} \frac{7t}{2}$$

$$\lim_{t \rightarrow 2} \frac{7 \lim_{t \rightarrow 2} t}{2}$$

$$\lim_{t \rightarrow 2} \frac{7}{2}$$

$$\frac{7(2)}{2} = 7$$

22) a)

$$(i) \lim_{x \rightarrow 2^+} [x] = \lim_{x \rightarrow 2^+} [-2] = \underline{-2}$$

$$(ii) \lim_{x \rightarrow 2} [x] = \lim_{x \rightarrow 2^-} [-2] = \underline{-2}$$

$$(iii) \lim_{x \rightarrow -2.4} [x] = \lim_{x \rightarrow -2.4} [-2.4] = \underline{-3}$$

b)

$$(i) \lim_{x \rightarrow n^-} [x] = \lim_{x \rightarrow n^-} [n-1] = \underline{n-1}$$

$$(ii) \lim_{x \rightarrow n^+} [x] = \lim_{x \rightarrow n^+} [n] = \underline{n}$$

c) ¿Para qué valores de  $a$  existe  $\lim_{x \rightarrow a} [x]$ ?

\* Para ninguno porque  $a$  no es un entero